verse. Brahmagupta established rules for dealing with positive numbers, which he called "fortunes," and negative numbers, which he called "debts," but his main achievement was his set of rules for the number zero. He showed that when zero is subtracted from or added to a number, that number does not change.

JABIR IBN HAYYAN
721–815 CE

Known in the West as Geber, Abu Musa Jabir ibn Hayyan al-Azdi was born in Tus (now Iran), studied in what is now Yemen, and worked as a physician and an alchemist in what is now Iraq. He is credited with a vast body of work on subjects that ranged from alchemy and astronomy to medicine and magic. Many of his alchemical writings were later translated into Latin and became influential in the evolution of chemistry in Europe. In these writings, ibn Hayyan emphasized the importance of practical experimentation and described many processes still used in modern chemistry, including distillation and crystallization. He died at the age of 94.

ABD AL-RAHMAN AL-SUFI
903–986 CE

Persian astronomer Abd al-Rahman al-Sufi made the first record of what are now known to be galaxies. In *The Book of Fixed Stars*, he described a "little cloud," now known as the Andromeda Galaxy, and also identified the Large Magellanic Cloud, which he called the White Ox. He could not have seen this from his home in Isfahan, in what is now central Iran, so he must have heard about it from astronomers in Yemen and sailors crossing the Arabian Sea. Al-Sufi's main work was translating Ptolemy's *Almagest* into Arabic.

AVICENNA
980–1037

Born near Bukhara, in present-day Uzbekistan, Avicenna (also known as ibn Sina) was a remarkable scholar who had memorized the Qur'an by the age of 10. At 16, he gained fame as a doctor when he cured the Samanid ruler Nuh ibn Mansur of a mystery illness. He was rewarded with the use of the ruler's extensive library, and produced the first of many works soon after. Avicenna traveled widely, working in royal courts across Asia. In philosophy, he is famous for his thought experiment the "Flying Man," with which he argued that the mind is distinct from the body. In 1025, he completed his major work, *The Canon of Medicine*, which explains how to diagnose and treat illnesses.

TROTA OF SALERNO
c.11th century

Trota of Salerno (a city in what is now southern Italy) was a female physician who lived and worked at a time when medical treatments offered to women were often barbaric. Trota approached this problem in a practical way, writing not only about what ought to work in the treatment of women's diseases, but also what did work in practice. Her most influential work was her contribution to the *Trotula*, a popular medical text in medieval Europe.

ROGER BACON
1214–1292

English friar Roger Bacon was a pioneer of experimental science who outlined his ideas about mathematics, physics, optics, and alchemy in his *Opus Majus* ("Great Work"), which he presented to Pope Clement IV as a plea for church reform. Unfortunately, the pope died before he could read the work, and Bacon was imprisoned by his fellow Franciscans for his radical views. His studies of light and how it interacts with objects heralded a new approach in optics. Bacon was also the first European to describe how to make gunpowder. His thirst for knowledge earned him the nickname "Doctor Mirabilis" (Wonderful Teacher).

CHOE MU-SEON
1325–1395

Born into a wealthy family, Choe Mu-Seon's ambition to bring the recipe for gunpowder to Korea was inspired by the fireworks he saw as a boy. Choe eventually achieved his aim by bribing a Chinese merchant to give him the secret recipe. Korea began to produce its own gunpowder soon after in c.1375. Choe went on to invent various firearms, which he put to the test against the pirates who threatened Korea's coastal regions, as well as the Japanese at the Battle of Jinpo in 1380.

ALI QUSHJI
1403–1474

Born in Samarkand, in what is now Uzbekistan, Ali Qushji studied astronomy and mathematics under the ruler and astronomer Ulugh Beg, who founded the Samarkand Observatory (of which Qushji later became director). There, Qushji wrote about the phases of the Moon and contributed to Beg's star catalog. After Beg was killed, Qushji moved to Constantinople (present-day Istanbul), where he founded a school for teaching traditional Islamic sciences. He is best known for his efforts to separate astronomy from philosophy and for providing empirical evidence that the Earth moves.